Renal Function — eGFR, CrCl & Urea:Creatinine + KDIGO CKD

adults and children · creatinine in µmol/L

eGFR / CKD

FORMULAS

CKD-EPI 2021 (adults, STG): `142 × min(Scr/κ,1)^α × max(Scr/κ,1)^−1.200 × 0.9938^age × 1.012 [F]` · Scr mg/dL = µmol/L ÷ 88.4; κ 0.7 F / 0.9 M; α −0.241 F / −0.302 M; race-free

MDRD (IDMS): `175 × Scr^−1.154 × age^−0.203 × 0.742 [F]` · Scr mg/dL; original × 1.212 if Black (not recommended)

Cockcroft-Gault: `(140 − age) × weight × 1.23 [M] or 1.04 [F] ÷ Scr` · Scr µmol/L, weight kg → CrCl mL/min; drug dosing

Schwartz (children, STG) ✚: `eCrCl = K × height (cm) ÷ creatinine (µmol/L)` · K: LBW <2.5 kg 30 · infant 0–8 mo 40 · girls 2–16 y 49 · boys 2–12 y 49 · boys 13–16 y 60

Urea:creatinine (SI): `urea (mmol/L) ÷ creatinine (µmol/L) × 1000` · >100 suggests prerenal / UGIB; <40 intrinsic renal (approximate)

BUN:Cr (US): `≈ SI ratio × 0.2475 (BUN mg/dL = urea mmol/L × 2.8)` · >20 prerenal

NOTES

- Use CKD-EPI for adults, Schwartz for children ✚
- eGFR invalid in AKI (non-steady state), extremes of muscle mass, pregnancy
- Schwartz K table has no value for 8 months–2 years; bedside Schwartz (2009) uses K = 36.5 for all ages ‡

KDIGO 2024 CKD HEAT MAP (FIGURE 7.1) ✚

eGFR	A1 <3	A2 3-30	A3 >30 mg/mmol
G1 ≥90	low	moderate	high
G2 60-89	low	moderate	high
G3a 45-59	moderate	high	REFER
G3b 30-44	high	REFER	REFER
G4 15-29	REFER	REFER	REFER
G5 <15	REFER	REFER	REFER

SA STG

- CKD = kidney damage or eGFR <60 for >3 months ✚
- Stages 0–3 can be managed at primary care once cause and plan are set; all stage 4–5 → specialist ✚
- PHC: refer G1–G3a with A2/A3, and all G3b or worse ‡ ✚
- STG prints UACR '30 mg/mol' — read mg/mmol ‡ ✚

HOSP STG & EML Hospital Level Adults, 6th ed. 2024 (updated Mar 2026) — pp 7.1, 7.2, 7.3 · **PHC** STG & EML Primary Healthcare, 8th ed. 2024 (updated Sept 2026) — pp 8.3 · **PAED** STG & EML Paediatric Hospital Level, 2025-2029 cycle (updated Mar 2026) — pp 6.15 · **CKDEPI** Inker LA et al. CKD-EPI 2021 (race-free). N Engl J Med 2021;385:1737-49 · **MDRD** Levey AS et al. MDRD (IDMS). Ann Intern Med 2006;145:247-54 · **CG** Cockcroft DW, Gault MH. Nephron 1976;16:31-41

✚ = SA STG / national guideline (pages listed above). Unmarked content: international source named above. ‡ = STG misprint or sources differ (see app). ANDH · renal · updated 2026-10-14